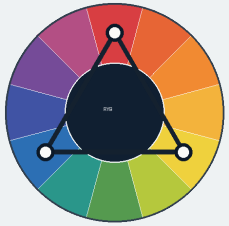
# COLOUR HARMONY

## A PROFESSIONAL REFERENCE FOR ARTISTS, VISUAL DESIGNERS AND 3D CREATORS

### HOW TO READ THIS GUIDE

The six topics in this guide are **colour harmony schemes** - practical ways of arranging hues within colour theory. They are not six unrelated theories. Each scheme describes a geometric relationship on a colour wheel, then relies on value, chroma, proportion, material and context to become a working design palette.

**Working principle:** choose a harmony to establish relationships, then build hierarchy. A mathematically tidy palette can still fail if the values merge, all colours compete, or lighting changes their appearance.



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### FOUNDATIONS THAT CHANGE THE RESULT

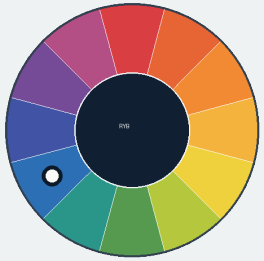
**Hue** names the colour family. **Chroma** describes colourfulness relative to a reference; **saturation** is model-dependent. **Value / lightness** describes perceived lightness. **Temperature** is relational: one red can be cooler than another red depending on its bias.
**Wheel models differ:** RYB is a traditional painter's teaching wheel; RGB describes additive light; CMYK is print-oriented subtractive colour.

### PRODUCTION NOTE FOR DIGITAL AND 3D WORK

The guide uses an RYB wheel for familiar art relationships; HEX values are approximate sRGB screen references, not pigment formulas. Display calibration, colour space, view transform, exposure, light spectrum, roughness and metallic response alter the result. Judge the final camera view under intended lighting. For precise adjustments, perceptual spaces such as CIELAB/LCh or OKLCH can help control lightness and chroma.

# MONOCHROMATIC

**ONE HUE / CONTROLLED VALUE AND CHROMA**



## DEFINITION AND DESIGN INTENT

A monochromatic harmony holds the hue family approximately constant while varying lightness (value) and chroma (saturation). Its sophistication comes from tonal structure, material differentiation and carefully controlled accents - not from using one flat colour repeatedly.

**Construction rule:** Keep the hue anchor stable; build separation through value, chroma, texture, reflectance and lighting. In a strict version, hue drift is minimal. A slight cool/warm shift in shadows or highlights can add depth, but it moves the palette toward a near-monochromatic scheme.

**Wheel relationship:** RYB wheel: one selected hue sector (blue). The palette extends along a tonal axis rather than across multiple hue sectors.

## PALETTE ARCHITECTURE / sRGB HEX REFERENCES

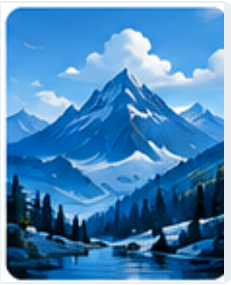
| Ink Navy | Deep Blue | Cobalt | Glacier | Ice |
|---|---|---|---|---|
| #10243D | #173F70 | #2D70B8 | #78AEE2 | #DDEBF7 |
| Shadow / anchor | Structural midtone | Primary material | Light plane | Highlight |

## THREE VISUAL APPLICATIONS



### CHARACTER / PROP

Use value steps to distinguish armour plates, recesses and specular edges.



### ENVIRONMENT

Atmospheric depth through progressively lighter, lower-chroma blue distance planes.



### PRODUCT / INTERIOR

A tonal family keeps ceramic forms quiet while gloss and silhouette carry interest.

## COMPOSITION AND CONTROL

**Build a value ladder.** Assign the darkest value to structural recesses and silhouette breaks, mid-values to the main mass, and light values to focal planes. Keep the brightest value scarce so the eye has somewhere to land. A useful palette check is to convert the design to greyscale: the form should remain legible without relying on hue alone.

**Separate materials without adding new hues.** Change roughness, gloss, transparency, texture scale and edge response. Blue anodised metal, matte polymer and translucent glass can share a hue family but still read as different materials.

**Where it performs well** Product families and restrained identity systems; interface states; premium packaging; atmospheric environment art; hero objects where silhouette and material finish should dominate. Monochromatic palettes are effective when the form itself carries the design.

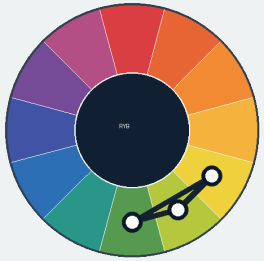
## FAILURE MODES / REFINEMENT

**Flat hierarchy:** widen the value range and reserve high-value highlights for focal areas.
**Colour-on-colour merging:** add edge contrast, texture or local value separation.
**Cold or sterile result:** introduce a controlled warm neutral in the background or lighting, rather than adding several competing hues.

**Studio exercise** Create a five-step material study using the same blue hue family. Make the silhouette, focal plane and shadow separation readable at thumbnail size before adding surface detail.

# ANALOGOUS

**NEIGHBOURING HUES / CONTROLLED TEMPERATURE FLOW**



## DEFINITION AND DESIGN INTENT

An analogous harmony groups neighbouring hue families on the chosen colour wheel. It creates a continuous colour flow because adjacent hues share visual relationships. The scheme can be narrow and quiet or broader and more expressive depending on the span selected.

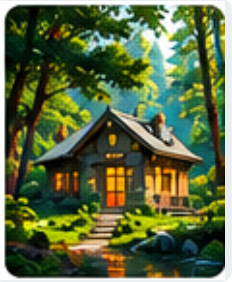
**Construction rule:** Choose a lead hue, one supporting neighbour and (optionally) a smaller accent neighbour. Two to four adjacent hue families are usually easier to manage than treating a large arc as equally important.

**Wheel relationship:** RYB wheel: yellow, yellow-green and green sectors form a warm-to-cool progression through a related hue region.

## PALETTE ARCHITECTURE / sRGB HEX REFERENCES

| Moss | Leaf | Fresh | Chartreuse | Lime Light |
|---|---|---|---|---|
| #315C32 | #4E8A3E | #82AE37 | #B8C83C | #D9D85A |
| Deep anchor | Lead green | Support | Accent | Highlight |

## THREE VISUAL APPLICATIONS



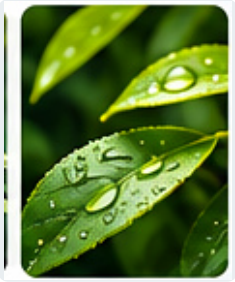
### ENVIRONMENT

A connected green-yellow range makes foliage, light and architecture feel part of one climate.



### CHARACTER / PROP

A green armour family feels cohesive; value and metallic response define separate components.



### MATERIAL DETAIL

Small yellow-green shifts add life to a botanical surface without breaking the palette.

## COMPOSITION AND CONTROL

**Establish a lead hue.** Let one colour carry the largest area or highest chromatic authority. Use its neighbours to turn surfaces, guide movement or separate adjacent design zones. Avoid giving every hue equal area and saturation; that removes the hierarchy that makes the scheme feel intentional.

**Use value to create focal contrast.** Analogous hues can have similar luminance. Separate focal objects from their surroundings with value, texture, edge definition or a deliberate shift in chroma - not just a small hue change.

**Where it performs well** Landscape and environment design; foliage and organic systems; wellness and sustainability identities; food packaging; natural product materials; scenes where a calm, immersive or continuous atmosphere is required.

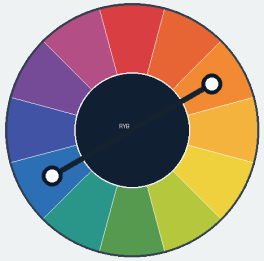
## FAILURE MODES / REFINEMENT

**Insufficient separation:** adjacent hues may merge when values are close. Push the focal hue lighter/darker or raise its chroma selectively.
**Unintentional rainbow effect:** reduce the number of equally prominent hues and define a clear lead/support/accent order.
**Muddy mixtures in paint:** choose cleaner pigments and test mixtures; adjacent does not automatically mean clean when pigments contain complex undertones.

**Studio exercise** Build an environment palette with one lead green, one yellow-green support and one darker green anchor. Test it at thumbnail scale and under both warm and cool lighting.

# COMPLEMENTARY

OPPOSING HUES / MAXIMUM HUE CONTRAST



## DEFINITION AND DESIGN INTENT

A complementary harmony pairs hues positioned opposite one another on a selected colour wheel. The relationship creates a strong perceptual distinction when the colours are placed next to each other, making it useful for focal emphasis, visual separation and energetic compositions.

**Construction rule:** Use one hue as the dominant field and its complement as a deliberate accent or counterweight. Complementary colours do not need to be equal in area, brightness or saturation; in most professional compositions they should not be.

**Wheel relationship:** RYB wheel: blue and orange occupy opposing sectors. The connecting line marks the complementary relationship.

## PALETTE ARCHITECTURE / sRGB HEX REFERENCES

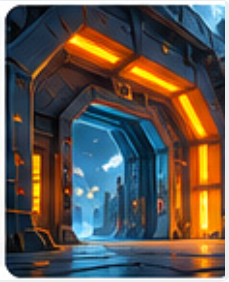
| Midnight | Ultramarine | Signal Blue | Amber | Warm Sand |
|---|---|---|---|---|
| #122743 | #285BA4 | #4387C9 | #EF8A2F | #F4C17A |
| Dominant field | Primary blue | Secondary blue | Accent / light | Warm highlight |

## THREE VISUAL APPLICATIONS



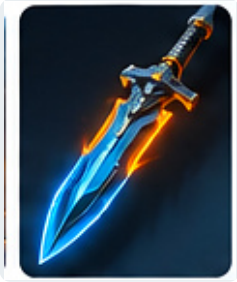
### CHARACTER / SCI-FI

Orange emissive details punctuate a cool blue armour mass and direct attention to the face and joints.



### ENVIRONMENT

Warm architectural lighting separates a subject from a cool blue structure and atmosphere.



### PROP / WEAPON

A restrained warm accent becomes a functional focal cue against a blue body.

## COMPOSITION AND CONTROL

**Control the contrast budget.** Let the dominant hue occupy the larger area; reserve the complement for focal points, lights, calls to action or edges that need to advance. If both hues are highly saturated and used in equal proportions, they compete rather than cooperate.

**Use neutral transitions.** Dark neutrals, warm greys or low-chroma intermediate colours can separate high-chroma complementary regions and reduce visual vibration. Value contrast and edge control still determine readability.

**Where it performs well** Action-oriented game art; sci-fi interfaces and props; sports and entertainment identities; safety or attention cues; product renders requiring a clear focal hierarchy. Blue-orange is common in cinematic lighting because cool shadows and warm highlights can separate subject from environment.

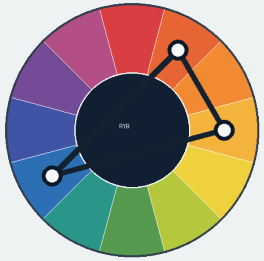
## FAILURE MODES / REFINEMENT

**Edge vibration:** avoid placing similarly bright, fully saturated complements in thin adjacent bands; lower chroma or add a value step.
**Overuse of accent:** reduce the complement's area and reserve it for meaningful information.
**False assumption:** complementary contrast is not the same as pigment mixing. Mixing complements often reduces chroma toward a neutral; visual adjacency creates the contrast effect.

**Studio exercise** Render the same object three times: 80/20, 60/40 and 50/50 area balance. Compare which version gives the strongest hierarchy without colour competition.

# SPLIT COMPLEMENTARY

ONE BASE HUE / TWO NEAR-COMPLEMENT ACCENTS



## DEFINITION AND DESIGN INTENT

A split-complementary harmony combines a base hue with the two hues adjacent to its direct complement. It retains a strong warm-cool or hue contrast while giving the designer two related accent directions instead of one opposing accent.

**Construction rule:** Choose the base hue first. Find its complement on the wheel, then select the two neighbours on either side of that complementary sector. Treat the base as the identity colour; give the two split accents different jobs, areas or saturation levels.

**Wheel relationship:** RYB wheel: blue is the base; red-orange and yellow-orange sit on either side of its complementary orange region.

## PALETTE ARCHITECTURE / sRGB HEX REFERENCES

| Abyss | Steel Blue | Rust | Saffron | Pale Sand |
|---|---|---|---|---|
| #15345F | #2F6FA9 | #D65C37 | #F0A83B | #F1D6B5 |
| Base / field | Support | Accent 1 | Accent 2 | Neutral |

## THREE VISUAL APPLICATIONS



### CHARACTER DESIGN

Blue remains the armour identity while two warm accents distinguish lighting and equipment.



### PRODUCT / PROP

An amber light source provides the primary accent; nearby red-orange elements create a secondary warm note.



### ENVIRONMENT

Autumn foliage and warm windows create two distinct accents against a blue-led structure and sky.

## COMPOSITION AND CONTROL

**Assign two distinct accent roles.** One split hue can carry illumination or energy; the other can identify a secondary object, warning, trim or background element. Keep their values and saturation differentiated so they do not merge into one warm mass.

**Preserve a dominant-cool / supporting-warm relationship.** If all three hue groups occupy similar areas, the palette becomes busy. Add a low-chroma neutral to create breathing room between the accents.

**Where it performs well** Character and costume design; props with multiple functional details; story-driven environments; editorial illustration; brand systems that need more than one accent but still require a clear base identity.

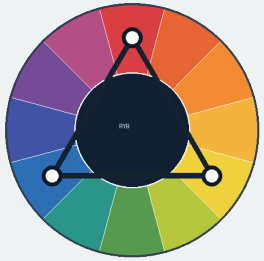
## FAILURE MODES / REFINEMENT

**Two accents compete:** choose one as primary and the other as secondary by scale, chroma or placement.
**Warm region becomes a block:** separate the accents with neutral materials or different value ranges.
**Scheme drifts into triadic:** keep the split hues close to the two sides of the complement, rather than selecting unrelated thirds of the wheel.

**Studio exercise** Make a three-colour composition from a single blue base and two split accents. Produce two versions: one where the yellow-orange is primary, and one where red-orange leads.

# TRIADIC

THREE EVENLY SPACED HUES / BALANCED VARIETY



## DEFINITION AND DESIGN INTENT

A triadic harmony uses three hues spaced at approximately equal intervals around a chosen colour wheel. On a traditional RYB teaching wheel, a familiar triad is red, yellow and blue. It offers a broad range of colour identities while retaining a repeatable geometric relationship.

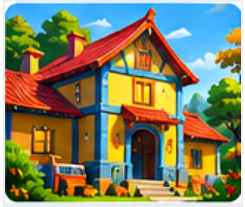
**Construction rule:** Select one lead hue, one support hue and one accent hue. Vary their value and chroma so the three hues do not shout with equal intensity. A triad can be vivid, but it does not require every colour to be bright or equally saturated.

**Wheel relationship:** RYB wheel: red, yellow and blue are placed at three evenly spaced positions, forming a triangle. RGB hue geometry produces a different triad; select the model intentionally.

## PALETTE ARCHITECTURE / sRGB HEX REFERENCES

| Oxide Red | Marigold | Royal Blue | Warm White | Dusty Rose |
|---|---|---|---|---|
| #B93645 | #E9BE43 | #2B5EA8 | #F0E6D4 | #D98D8A |
| Lead / warm mass | Highlight / accent | Cool support | Neutral | Soft bridge |

## THREE VISUAL APPLICATIONS



### ENVIRONMENT

Red, yellow and blue appear across architecture but are controlled by scale and value.



### CHARACTER / MECH

The lead colour defines the silhouette; the other two identify armour modules and focal details.



### PRODUCT / ASSET

A warm/cool triad separates functional panels while neutral areas preserve legibility.

## COMPOSITION AND CONTROL

**Build unequal emphasis.** Decide which hue owns the largest area or strongest value contrast. Let the second colour support form and the third mark focal details. Use neutral surfaces or lower-chroma versions to keep the palette from turning into a flat primary-colour scheme.

**Keep the value structure coherent.** The three hue groups can have different luminance. Check the composition in greyscale and make sure focal areas remain distinct from supporting masses.

**Where it performs well** Stylized environments; toy and consumer product concepts; character and mech design; packaging; posters and editorial work that need an expressive but structured palette.

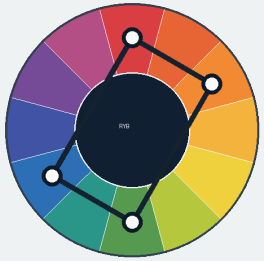
## FAILURE MODES / REFINEMENT

**All hues compete:** reduce the chroma of the support and accent colours.
**Primary-colour cliché:** move away from fully saturated pure hues; use complex, material-specific versions.
**Inconsistent mood:** unify the palette with shared neutrals, common lighting and consistent shadow colour.

**Studio exercise** Design a compact prop using all three hues, then make one hue dominant, one secondary and one accent. Compare it with an equal-area version to see how hierarchy changes.

# TETRADIC (DOUBLE COMPLEMENTARY)

FOUR HUES / TWO COMPLEMENTARY PAIRS



## DEFINITION AND DESIGN INTENT

A tetradic harmony uses four hues arranged as two complementary pairs. The hues can be organised as a rectangle or square on the chosen colour wheel. It offers a broad colour vocabulary, but the additional relationships increase the need for hierarchy and controlled chroma.

**Construction rule:** Choose a dominant complementary pair first. Add the second pair as a supporting colour family, not as a second equally loud system. The palette usually works best when one hue leads, one supports, and the remaining two appear as controlled accents.

**Wheel relationship:** RYB wheel: two opposing pairs are selected - red with green, and orange with blue. The four points form a rectangle-like relationship on the teaching wheel.

## PALETTE ARCHITECTURE / sRGB HEX REFERENCES

| Garnet | Amber | Jade | Royal Blue | Stone |
|---|---|---|---|---|
| #B8444F | #E99C39 | #3D855D | #2A5A9A | #E9DEC9 |
| Warm accent | Warm support | Cool / natural support | Dominant cool anchor | Neutral |

## THREE VISUAL APPLICATIONS



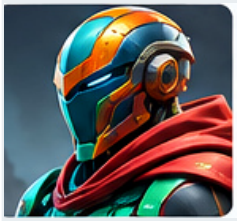
### ENVIRONMENT

Four hue families can describe architecture, foliage, banners and light while a dominant pair leads.



### INTERIOR / PRODUCT

A strong blue-green base is balanced by warm red and gold details, with neutral materials connecting them.



### CHARACTER DESIGN

The main cool body colour remains readable while red and amber accents distribute attention.

## COMPOSITION AND CONTROL

**Choose the leading pair before the second pair.** Establish which complementary relationship carries the main scene. Let the second pair distribute secondary objects, environment zones or material details. If every hue receives equal area and saturation, there is no visual entry point.

**Unify through value and context.** Repeat neutral materials, maintain a consistent shadow strategy and use chroma selectively. Four hue families can coexist when their value ranges and spatial roles are clearly differentiated.

**Where it performs well** Complex game environments; fantasy architecture and interiors; multi-product or multi-category brand systems; scenes with several focal objects; richly coloured illustration where the palette must support narrative complexity.

## FAILURE MODES / REFINEMENT

**Visual competition:** reduce the area or chroma of the second pair.
**Colour fragmentation:** repeat selected colours across objects to create rhythm and cohesion.
**Muddy or overdesigned output:** use neutral surfaces to connect colour groups and remove accents that do not serve a focal or functional purpose.

**Studio exercise** Create a four-hue environment palette. First block the scene using only the dominant pair; then introduce the second pair in two controlled locations and compare the focal hierarchy.